

Analysis PhD Exam, May 2000

Be sure to write all proofs in a neat and logical fashion. Give reasons for all steps!

1. State and prove the Vitali Convergence Theorem for integrals.
2. State and prove the Martingale Convergence Theorem.
3. Let $(X, \mathcal{B}, \mu)$ and $(Y, \mathcal{J}, \nu)$ be finite measure spaces. Construct $\mu \otimes \nu$ on $\mathcal{B} \otimes \mathcal{J}$. Give details.
4. Let $\{\mu_m\}$ be a sequence of real valued signed measures on (S, Σ) . Define

$$\lambda(A) = \sum_{m=1}^{\infty} \frac{1}{2^m} \frac{|\mu_m|(A)}{1 + |\mu_m|(S)}, \quad A \in \Sigma.$$

Show λ is a finite measure on Σ and $\mu_m \xrightarrow{\varepsilon-\delta} \lambda$ for each m .

5. Let $f : [0, b] \rightarrow \mathbb{R}$ be Lebesgue integrable. Suppose $\int_{[0, c]} f \, dm = 0$ for each $c \in [0, b]$. Show $f = 0$ a.e. m .
6. Let $(\Omega, \mathcal{F}, P)$ be a pr. space, Y an integrable r.v. and $\mathcal{E}$ a sub σ -field of $\mathcal{F}$. Suppose $\mathcal{E}$ and $\mathcal{F}(Y)$ are independent. What is $E(Y \mid \mathcal{E})$? Prove.
7. Let $\mathcal{X}$ be a B-space and suppose $\mathcal{Y}$ and $\mathcal{Z}$ are closed linear subspaces of $\mathcal{X}$ such that each $x \in \mathcal{X}$ has a unique representation of the form $x = y + z$, where $y \in \mathcal{Y}$ and $z \in \mathcal{Z}$. Show that there is a constant K such that $|y| \leq K|x|$ and $|z| \leq K|x|$, for each $x \in \mathcal{X}$, where y & z are given as above.

Hint: define a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ which is useful, use the closed graph theorem to get continuity; then define $S : \mathcal{X} \rightarrow \mathcal{Z}$, etc.